Cricket Valley Energy

Cricket Valley Transmission Line and Re-conductoring Project

Exhibit 5

Design Drawings

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In connection with the redacted Figures, Cricket Valley will consult shortly with Con Edison to coordinate a request for possible Critical Infrastructure Information protection for said figures.

EXHIBIT 5 DESIGN DRAWINGS

5.1 Introduction

Cricket Valley Energy Center, LLC ("Cricket Valley") is proposing to: (1) develop a new approximately 14.6-mile 345 kilovolt (kV) transmission line parallel to the existing Consolidated Edison Company of New York, Inc.'s ("Con Edison") 345 kV Transmission Line 398 ("Line 398") from the planned Cricket Valley switchyard (the "Cricket Valley Switchyard") in the town of Dover, New York to Con Edison's Pleasant Valley Substation in the town of Pleasant Valley, New York (the "Transmission Line"); and (2) re-conductor an approximately 3.4-mile segment of the existing 345 kV Transmission Line 398 in the town of Dover between the Cricket Valley Switchyard and the New York-Connecticut state line (the "Re-conductoring Segment") (collectively the "Project").

The Project will also include improvements to Consolidated Edison's Pleasant Valley Substation. New protection and communication system upgrades will be required within the existing control buildings at the Pleasant Valley Substation.

The Project right-of-way crosses through the towns of Dover, Union Vale, La Grange, and Pleasant Valley, all located in Dutchess County, New York.

5.2 Design Drawings

A detailed description of the proposed Project transmission structures for the Transmission Line and Re-conductoring Segment, including material of construction, color and finish, is included in Exhibit E-1, Description of Proposed Transmission Facilities and will be included in the EM&CP. A description of the proposed modification to the Pleasant Valley Substation and existing Line 398 is provided in Exhibit E-2.

Six figures are included with Exhibit 5 to demonstrate both typical and site specific design and materials standards for critical components of the Project.

Figure 5-1 provides a plan view of the layout of the interconnection of Cricket Valley to Line 398.

Figure 5-2 is a typical cross-section diagram of the proposed Transmission Line, illustrating the length, width, and height of the proposed and existing structures.

Figures 5-3a and 5-3b will provide depictions of Con Edison's existing Pleasant Valley Substation with upgrades required for the construction and operation of the Project superimposed. As noted above, in connection with these redacted Figures, Cricket Valley will consult shortly with Con Edison to coordinate a request for possible Critical Infrastructure Information protection for said figures.

Figures 5-4a and 5-4b are typical details of the proposed insulator assembly proposed for use by the Project.

Figure 5-5 provides typical details of the pole designs for the proposed structures.

Cricket Valley Energy Center LLC

Cricket Valley Transmission Line and Re-conductoring Project

Exhibit 5 – Figures

Figure 5-1 Cricket Valley Energy Center Tie In

Figure 5-2 Typical Cross Section 345 kV L-LINE and PV-CV Transmission Project

Figure 5-3a Con Edison Pleasant Valley 345 kV Switch Station REDACTED

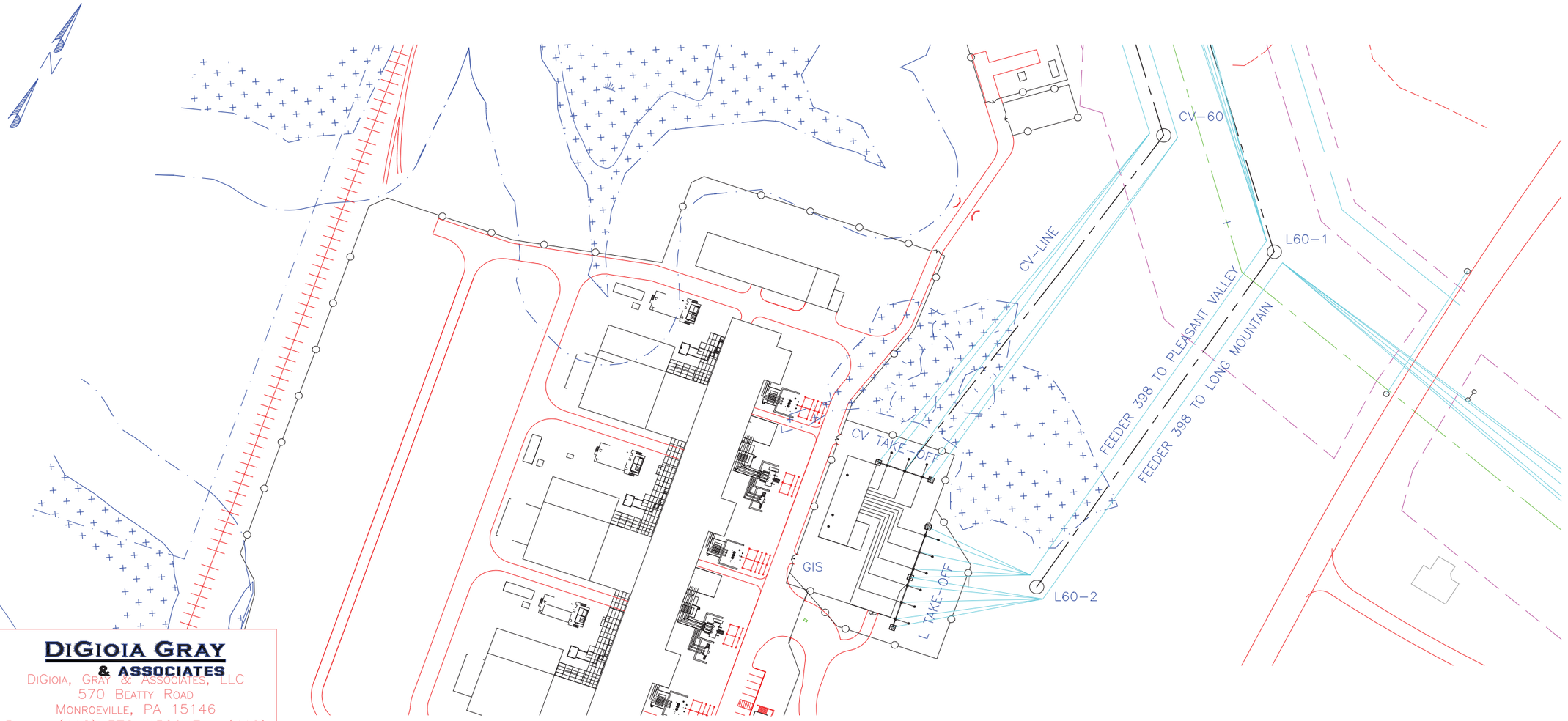
Figure 5-3b Con Edison Pleasant Valley Bay 2, RN2 Breaker REDACTED

Figure 5-4a Insulators - Standard V-String Suspension

Figure 5-4b Insulators - Standard Strain Assembly

Figure 5-5 Preliminary Pole Design

As stated above, in connection with the redacted figures, Cricket Valley will consult with Con Edison shortly to coordinate a request for possible Critical Infrastructure Information protection.



DRWN. BY:	MPS
DSGN. BY:	AVZ
CK'D. BY:	PGC
APP. BY: PGC	XX/XX/XXXX
DGA PROJ. NO.	2013-195

FRACTIONAL 0 1 2 — DECIMAL 2 1 0

DiGioia Gray & Associates

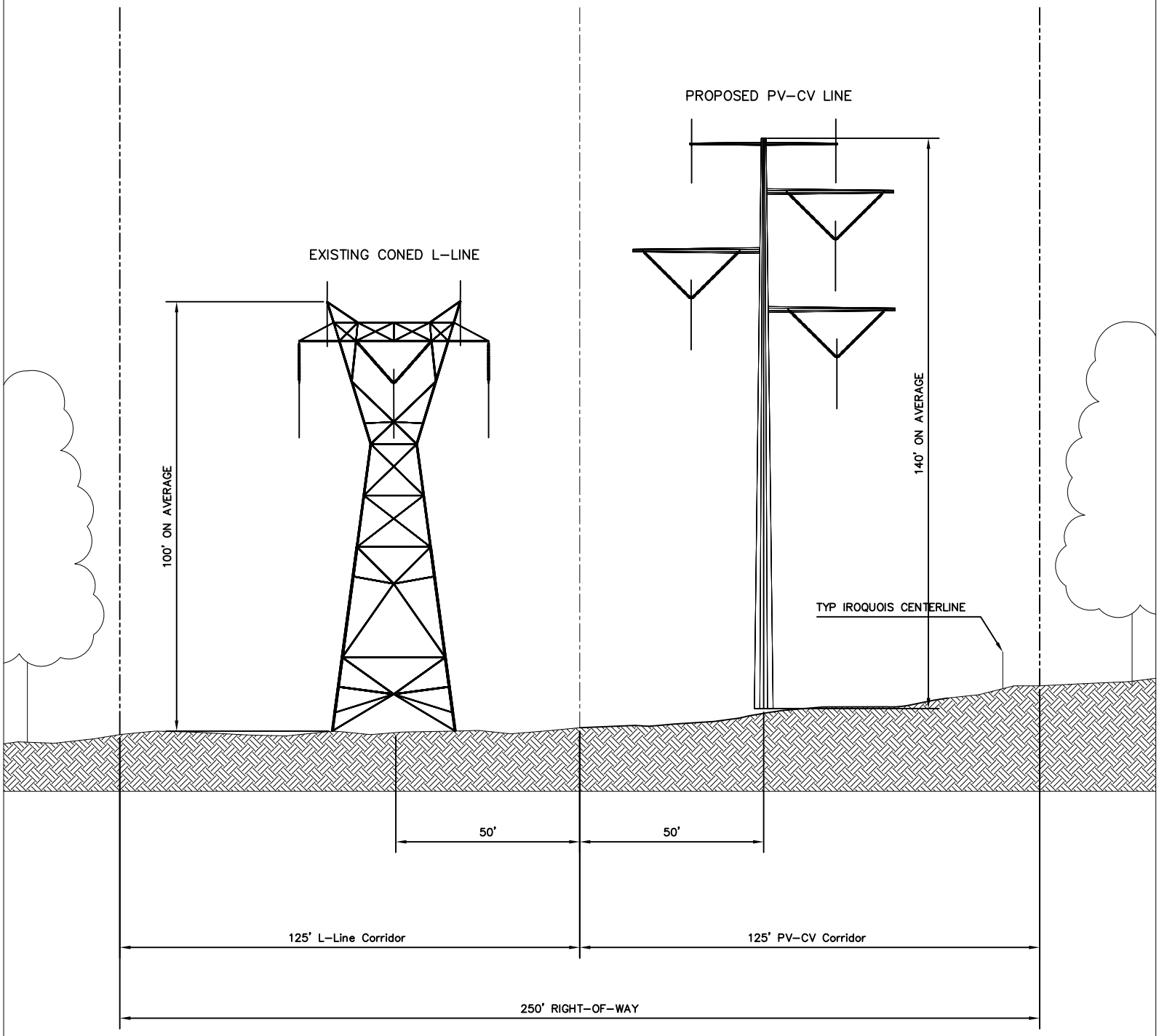
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GENERAL NOTES:

1. PROPERTY AND R/W DATA FROM HAND DRAWN VERSION.
2. PLAN AND PROFILE DRAWING GENERATED FROM PLS–CADD MODEL.

100.0 ft. Horiz. Scale

40.0 ft. Vert. Scale

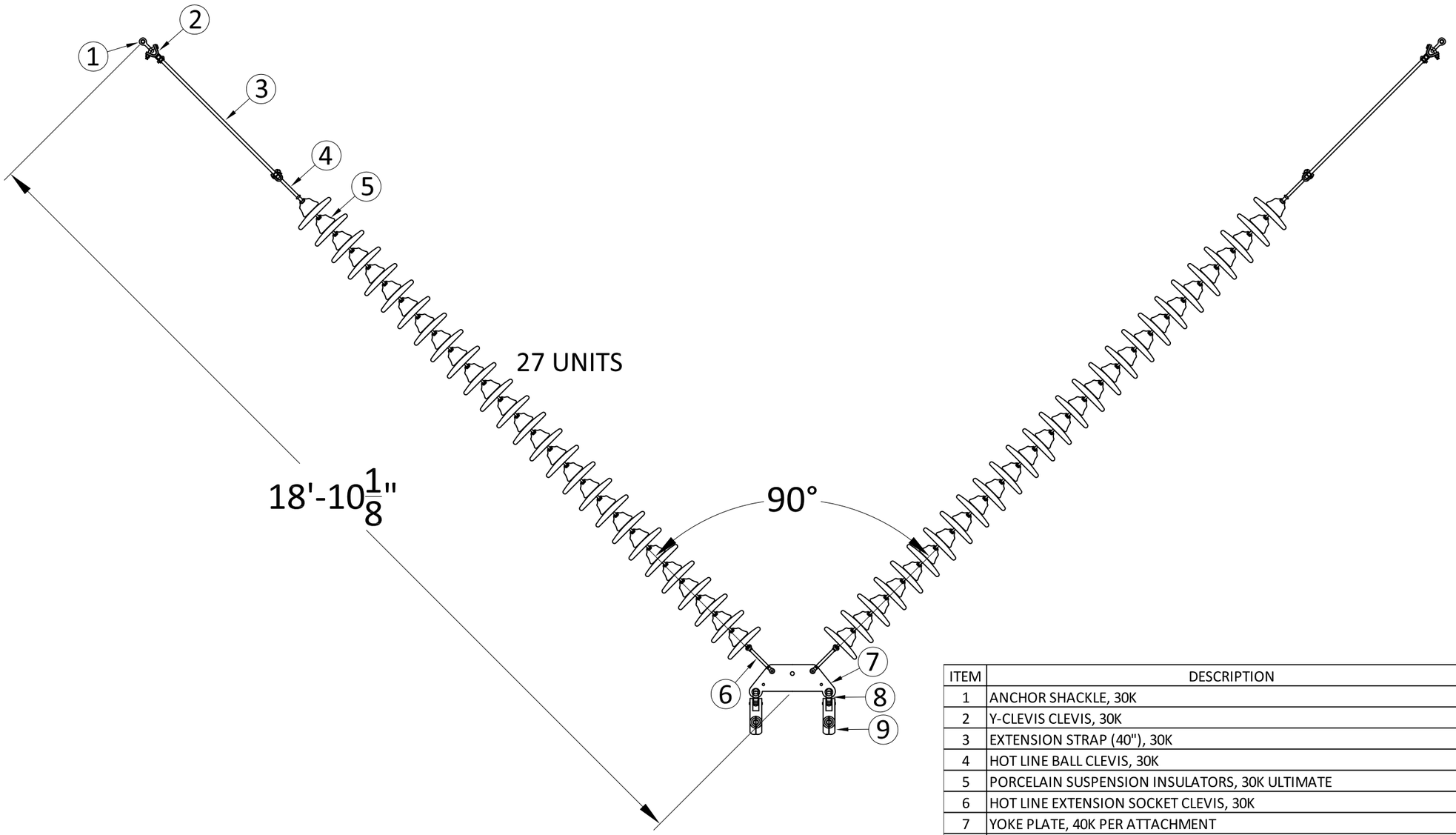


TYPICAL CROSS SECTION
L–LINE & PV–CV LINE 345KV

Figure 5-2

DiGioia, Gray & Associates, LLC

TYPICAL CROSS SECTION
345KV
L–LINE & PV–CV 345kV LINE
TRANSMISSION PROJECT

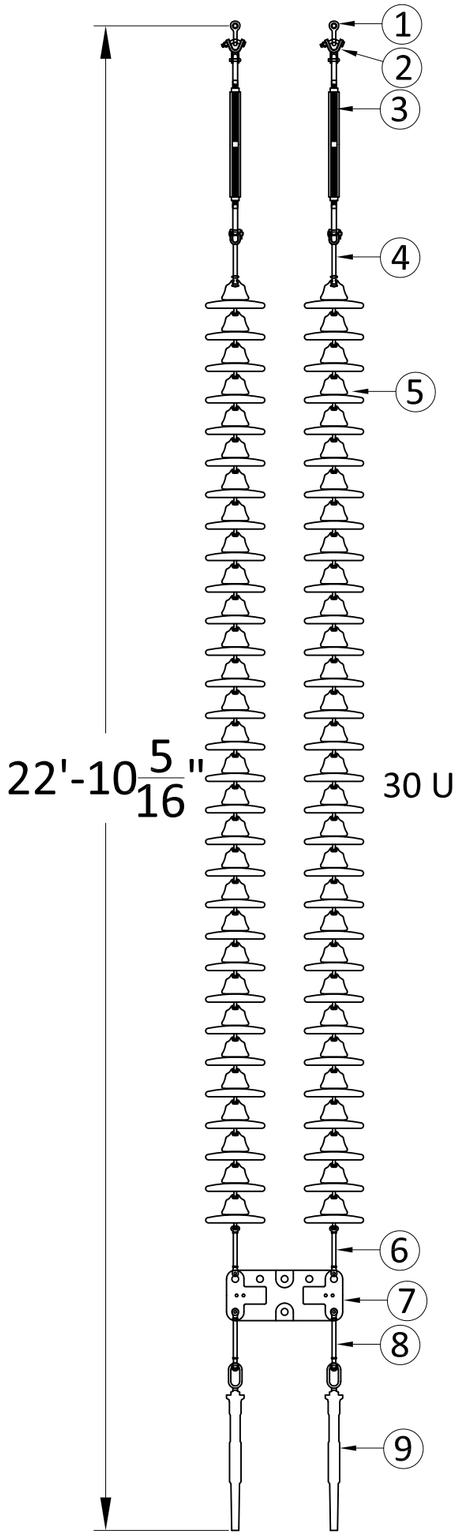


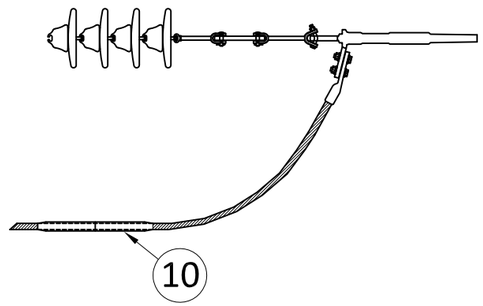
ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	ANCHOR SHACKLE, 30K	HUBBELL	AS25WBNK	2
2	Y-CLEVIS CLEVIS, 30K	HUBBELL	YCC30	2
3	EXTENSION STRAP (40"), 30K	HUBBELL	ES30781940	2
4	HOT LINE BALL CLEVIS, 30K	HUBBELL	HBC30	2
5	PORCELAIN SUSPENSION INSULATORS, 30K ULTIMATE	LAPP	5960A-70	54
6	HOT LINE EXTENSION SOCKET CLEVIS, 30K	HUBBELL	HSC100	2
7	YOKE PLATE, 40K PER ATTACHMENT	HUBBELL	930064004	1
8	STANDARD CLEVIS EYE, 30K	AFL	SCE-55-1375	2
9	HIBUS, CONDUCTOR SUSPENSION CLAMP, 25K	AFL	HCSHT1116	2

30K ULTIMATE CAPACITY PER STRING
15K WORKING CAPACITY PER STRING
WEIGHT OF ASSEMBLY 867.69 LB

STANDARD V-STRING SUSPENSION
(0 TO 3 DEG)

Figure 5-4a



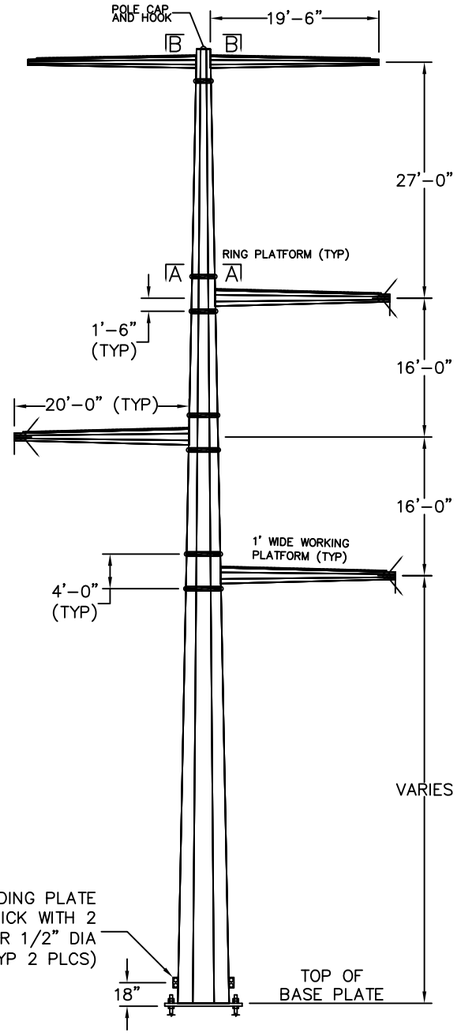


ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	ANCHOR SHACKLE, 30K	HUBBELL	AS25WBNK	2
2	Y-CLEVIS CLEVIS, 30K	HUBBELL	YCC30	2
3	TURNBUCKLE (32"), 36K	HUBBELL	TB78EE18	2
4	HOT LINE BALL CLEVIS, 50K	HUBBELL	HBC50	2
5	PORCELAIN SUSPENSION INSULATORS, 40K ULTIMATE	LAPP	61236H-70	60
6	HOT LINE EXTENSION SOCKET CLEVIS, 50K	HUBBELL	HSC110501	2
7	YOKE PLATE, 60K PER ATTACHMENT	HUBBELL	YPR6019401	1
8	HOT LINE Y-CLEVIS CLEVIS, 30K	HUBBELL	HYCC30	2
9	HI TEMP COMPRESSION DEAD END ASSEMBLY (MALLARD ACSS)	AFL	E33143HTEHV	2
10	HI TEMP JUMPER CONNECTOR (MALLARD ACSS)	AFL	5030.122HT	1

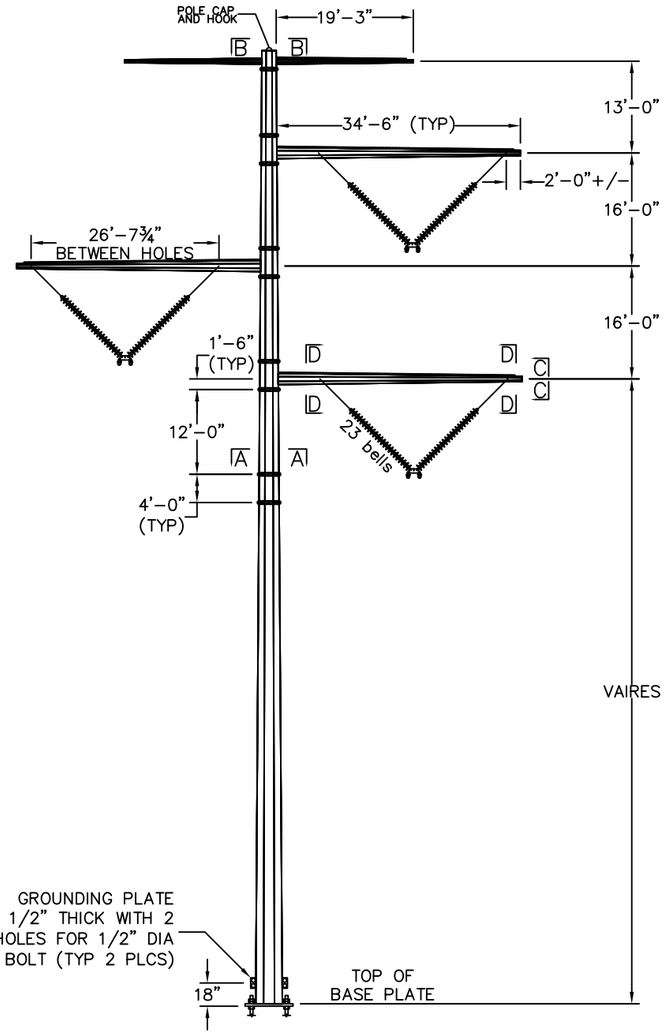
60K ULTIMATE CAPACITY PER ASSEMBLY
40K WORKING CAPACITY PER ASSEMBLY
WEIGHT OF ASSEMBLY 1116.725 LB

STANDARD STRAIN ASSEMBLY

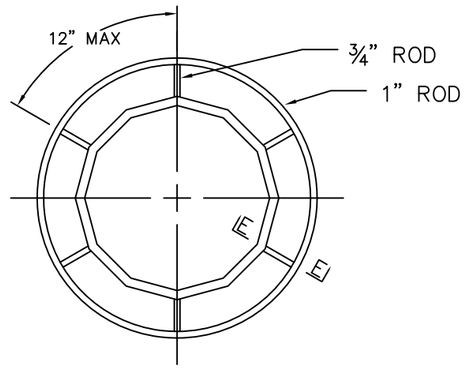
Figure 5-4b



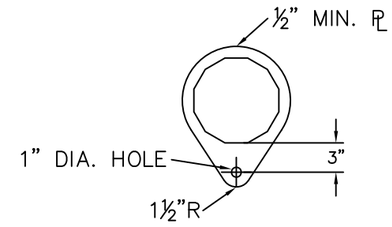
TYPICAL STRAIN & DEAD END
POLE STRUCTURE



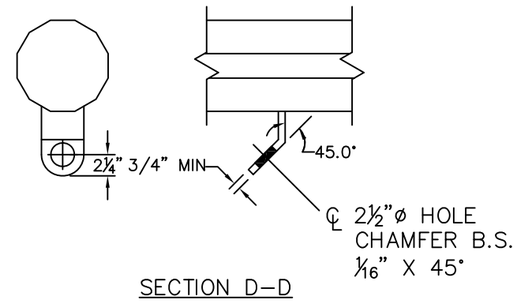
TYPICAL V-STRING
POLE STRUCTURE



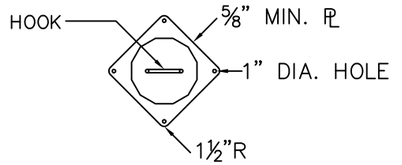
SECTION A-A
RING PLATFORM



SECTION C-C
ARM END PLATE

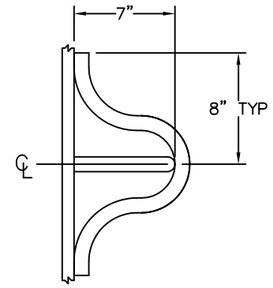


SECTION D-D



HOOK DIMENSION TO BE DETERMINED

SECTION B-B
POLE CAP



SECTION E-E

NOTES:
1. LIMIT POLE SECTION WEIGHTS TO 18,000# OR LESS.
2. UTILIZE LAP JOINTS BETWEEN SECTIONS MODIFIED FOR AIR CRANE ERECTION.
3. LADDER CLIPS, VANGS AND SIGNAGE ON POLES TO BE DETERMINED.
4. LIMIT POLE DEFLECTION TO 1.5% OF POLE HEIGHT UNDER NORMAL LOADING CONDITIONS.
5. STEEL POLE SHALL BE GALVANIZED IN ACCORDANCE WITH ASTM A123.
6. ANCHOR BOLT LENGTH AND END CONDITIONS DEPENDS ON FOUNDATION TYPE

PRELIMINARY

DRAWN BY:	MJS
DESIGNED BY:	PGC
CHECKED BY:	AVZ
APP. BY: PGC	1/8/13
DGA PROJ. NO.:	2012-184

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Figure 5-5

PLEASANT VALLEY TO CVEC 345kV
V-STRING, STRAIN & DEAD END
POLE STRUCTURES